

ENGN/PHYS 1590 Final Report:

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Methodology and Fabrication Process

Our team followed a pn-junction diode fabrication process using n-type silicon wafers and a Boron dopant. The fabrication sequence involved multiple steps performed in the clean room.

Wafer Preparation and Cleaning

We began with a 2-inch n-type silicon wafer. The wafer was first cleaned using a solvent clean procedure to remove organic contaminants. The cleaning sequence followed acetone/isopropanol/DI water process. Care was taken to prevent the wafer from drying between solvent steps to avoid residue formation. (see Image 1)

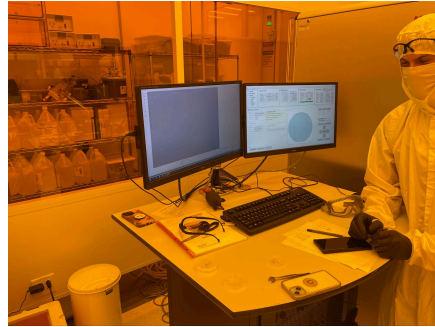


Image 1

Lithography for Diode Windows

Our first lithography step defined the regions for the pn junction formation. We used the MLA-150 maskless lithography tool (shown in image 2), which uses a UV laser to directly write patterns onto the photoresist-coated wafer.

The process involved:

1. Spinning photoresist at 4000 rpm for 30 seconds
2. Baking the wafer at 125°C for 30 seconds
3. Pattern exposure using the MLA-150 maskless aligner



Image 2

The lithography produced well-defined square patterns with clear edges. Since we used a negative photoresist, These squares would be the windows through which the dopant would diffuse into the silicon substrate after the oxide layer was etched.

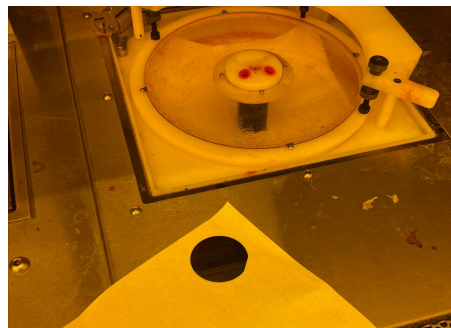


Image 3

Oxide Window Etching

Following lithography, we etched the exposed SiO_2 in the patterned windows using buffered oxide etchant (BOE). The etching process included:

1. Immersing the wafer in BOE solution such that the oxide was completely removed in the

- exposed areas (you can tell since silicon is hydrophobic while SiO₂ is hydrophilic)
2. Thorough rinsing with DI water and N₂ drying
 3. Photoresist removal using acetone, followed by cleaning in isopropanol and DI water

Doping Process

For the doping step, we:

1. Spin-coated the Boron spin-on dopant (SOD) at 3000 rpm for 30 seconds (see image 3 for centrifuge)
2. Baked the wafer for 30 seconds at 125°C
3. Loaded the wafer into the furnace for dopant drive-in

The furnace anneal was conducted at approximately 1000°C for 100 minutes with inert gas flow to prevent atmospheric contamination. This process likely created a diffusion depth of approximately 1-2µm, forming the pn junction beneath the oxide windows. Curiously, under the microscope, we noticed a residue that would not come off. We believe this to be from the spin on dopant, ie. some of the glass shards that carried the boron stubbornly stayed on the chip. (see image 4 on the right)

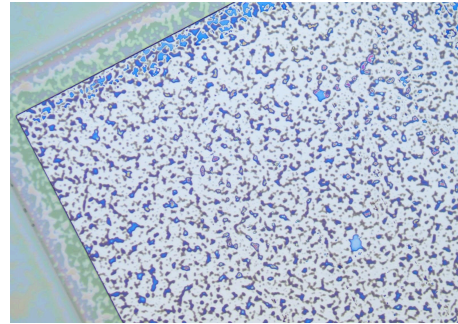


Image 4

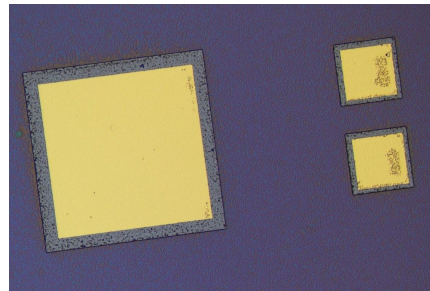


Image 5

Metallization

After the doping and junction formation:

1. We performed a second lithography step to define the metal contact areas, aligning precisely with the doped regions
2. Our first alignment did not work, so we had to redo it.
3. We deposited a gold contact using electron-beam evaporation
4. A lift-off process (acetone bath) removed the photoresist and excess metal, leaving contacts only in the desired regions

Image 5 shows three completed devices.

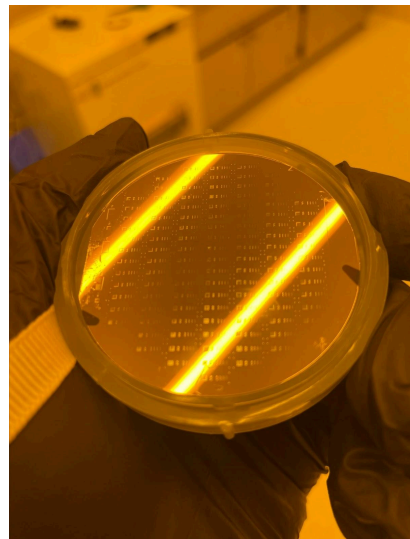


Image 6

Final Processing

The back contact was formed by applying indium metal to the substrate side of the silicon wafer.

This large-area contact ensured good electrical connection to the base layer of the pn junction structure.

The completed devices, as shown in Image 5, feature square metal contacts of various dimensions. The yellow areas represent the gold contact pads, while the surrounding gray borders are the isolation regions.

Measurement Setup

We used a probe station equipped with micromanipulators and fine-tipped needles to make electrical contacts to the gold pads on our devices. The probe station was connected to a voltage device that allowed the researchers to sweep between selected voltages and gather current measurements, as discussed below.

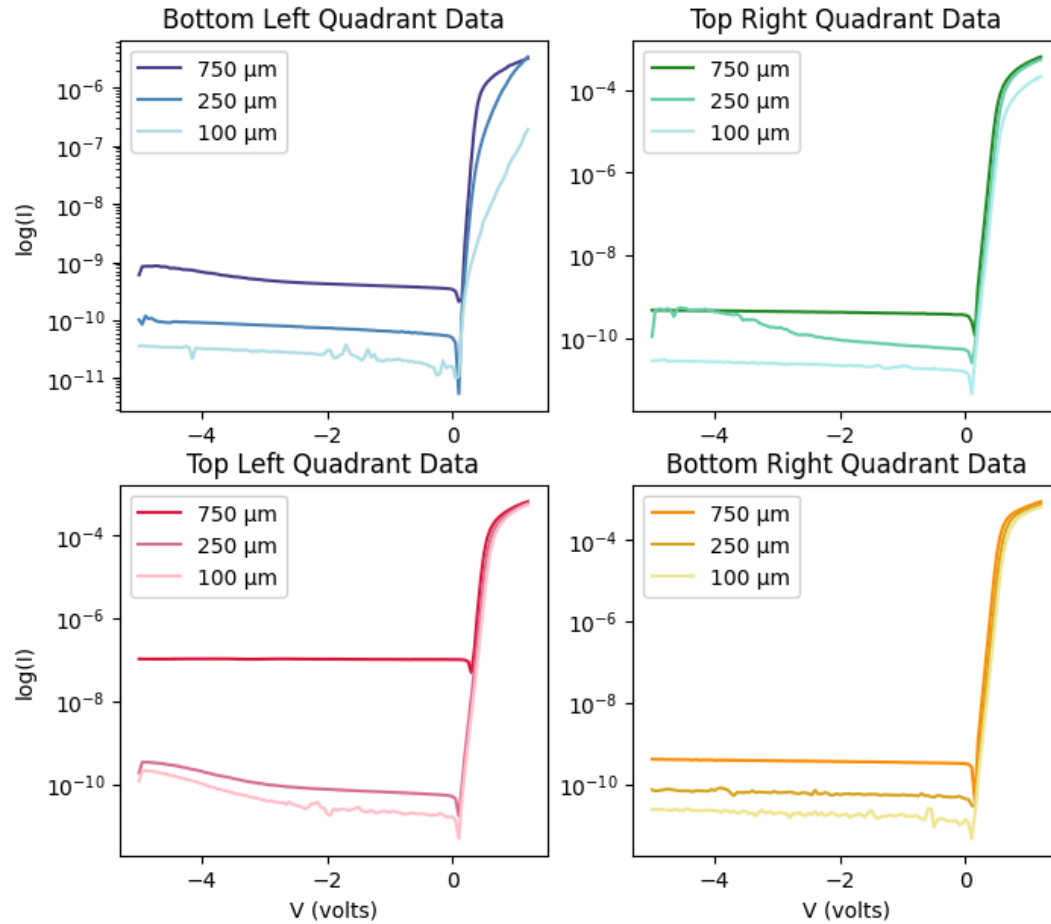
The testing configuration consisted of:

1. One probe tip contacting the gold pad on top of the p-type region
2. Another connection to the indium back contact for the n-type substrate
3. Configuration of the analyzer to sweep voltage while measuring current

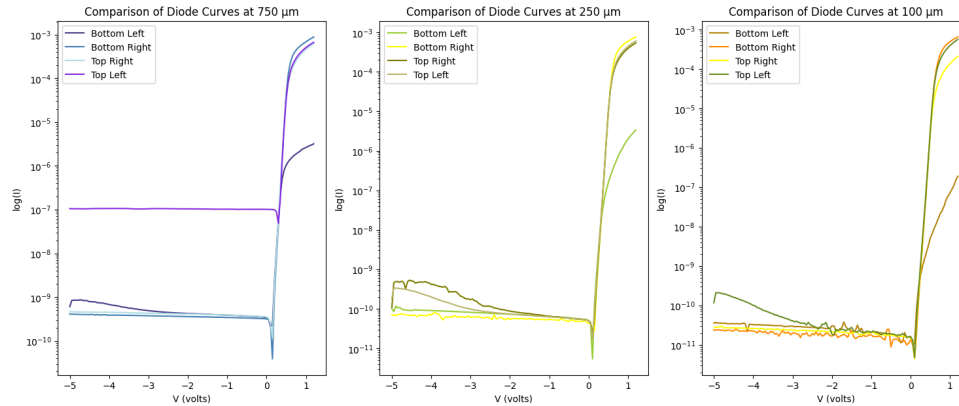
Results and Discussion

I-V Characteristics

We measured the I-V characteristics of diodes for all four quadrants of the wafer and for three different diode sizes (100 μm , 250 μm , 750 μm). This data is plotted below:



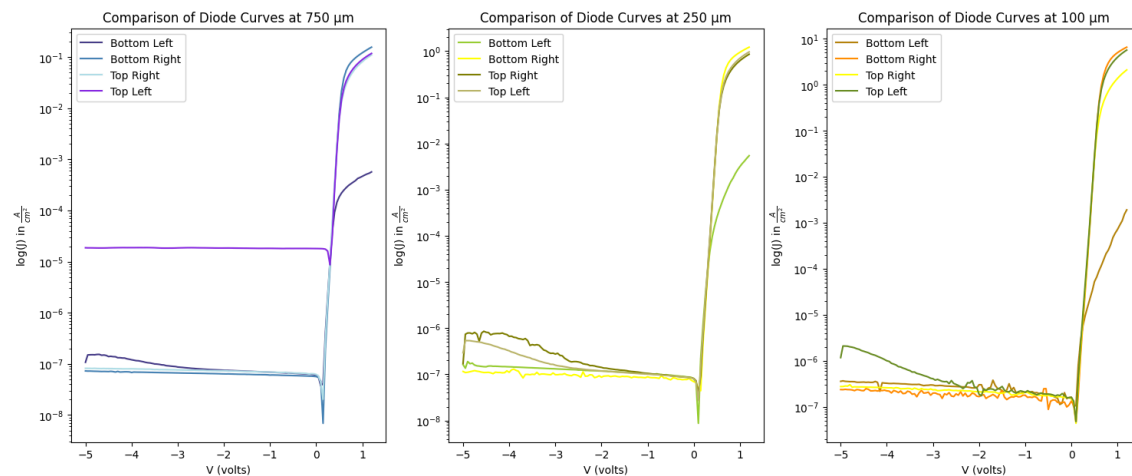
All diodes exhibited an exponential increase in current when under forward bias, consistent with our expectations. The current is proportional to the size of the diode as well which is similarly expected. Most diodes also showed minimal leakage (a less noisy, flatter line) when under reverse bias. However, the 100 and 250 μm diodes consistently displayed more noise than the 750 μm diode. This may be due to the fact that the leakage was somewhat absolute, rather than proportional to the current, causing the noise to be more apparent when the current was smaller. It also may be due to the fact that leakage occurs at defects on the edges of the junction perimeter and the perimeter-to-area ratio is naturally higher in the smaller diodes. We can see this same data graphed by diode size here:



Similarly, all quadrants are fairly similar except for the Bottom Left. This quadrant not only has more noisy data, but also displays non ohmic behavior around 10^{-6} Amps. This clearly indicates that the Bottom Left diodes are of significantly worse quality than those in other quadrants. This may be due to a number of errors in fabrication or testing including the incomplete removal of oxide or the probe not having a perfectly clean contact.

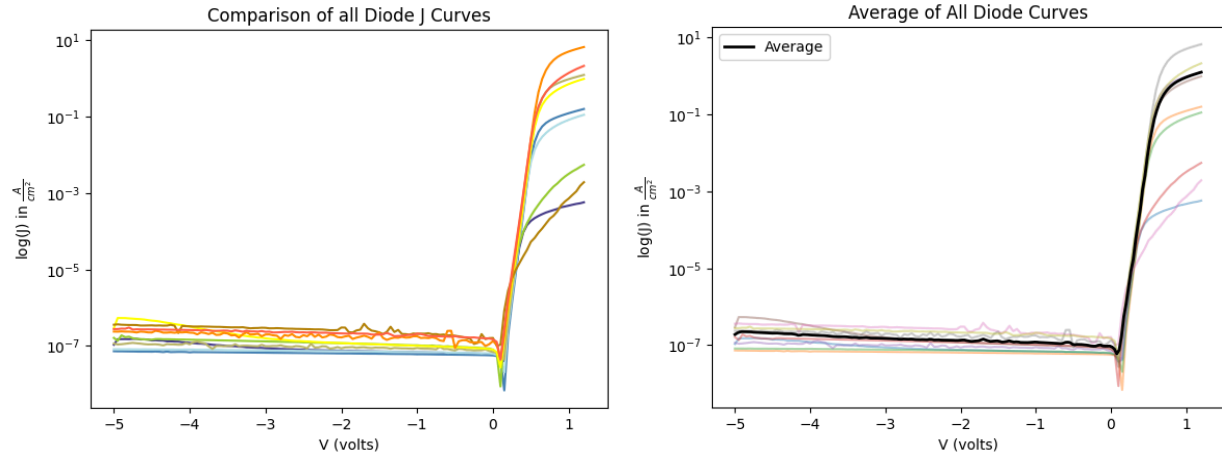
Ideal Diode Characteristics

To compare all the diodes, we calculated current density $J = I/A$, where A is the junction area. The plots of this data can be seen below.



These graphs are roughly similar as expected. The 100 and 250 μm diodes did display slightly higher current densities compared to the 750 μm diodes. As stated before, this may be partly due to the increased perimeter-to-area ratio in smaller devices, which enhances edge-related leakage. Another possibility is slight over-etching during fabrication, which would increase the actual junction area. This effect would be more noticeable in smaller diodes, aligning with the trend seen in our data.

Below we see all the J-V plots on one graph and an average of these plots:



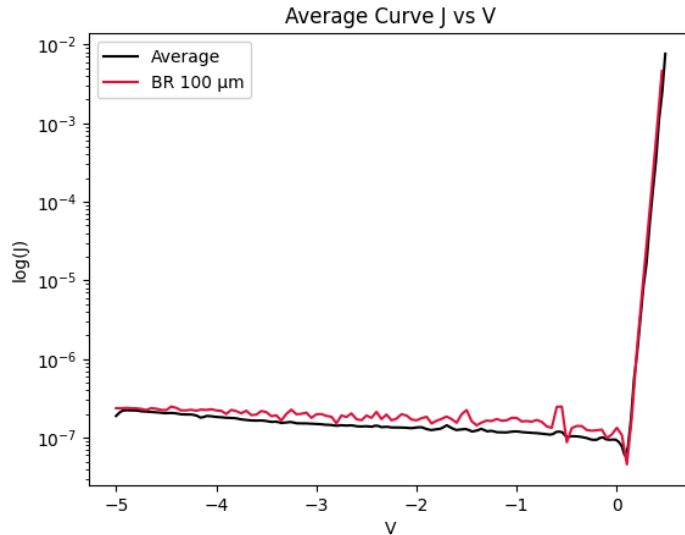
We see here how the majority of the diodes in reverse bias have a current density between 10^{-7} and 10^{-6} . In fact, we find that the average J_0 here is $\sim 1.4884 \times 10^{-7}$. This value aligns with what we expect for a silicon pn junction and matches the typical reverse saturation current density seen in moderately doped devices.

However, this value may be an overestimation. As discussed earlier, edge leakage most likely increased the reverse current. Many of the diodes also exhibited significant noise, which raised the reverse current density line as well. Additionally, it's important to note here that the average curve is particularly undefined around 0, instead of a clear transition from reverse to forward bias. This is likely due to photovoltaic effects, where ambient light reaches the diodes and generates electron-hole pairs. This shifts the current slightly towards the reverse-bias direction at zero bias.

Given that our constant $J_0 = q \left[\sqrt{\frac{D_p}{\tau_p}} p_n + \sqrt{\frac{D_n}{\tau_n}} n_p \right]$, we might go on to further solve for minority concentrations in each side of the junction. However, without further knowledge of precise doping in the n and p+ regions, it is difficult to estimate the mobility and diffusivity of our carriers. Hence, the researchers determined that further analysis over this constant would result in more speculation as opposed to definitive points of data.

Ideality factor for the Average Diode

In order to quantify the ideality of our diodes, we calculated the ideality factor (n) from the forward-bias slope of the average diode curve.



The diode equation is:

$$J = J_0(e^{\frac{qV}{nkT}} - 1)$$

Taking the natural log in the forward-biases region, we see that:

$$\ln(J) \approx \ln(J_0) + \frac{qV}{nkT}$$

From our plot of $\ln J$ vs V , we can get the slope $\frac{qV}{nkT}$, from which we can get n .

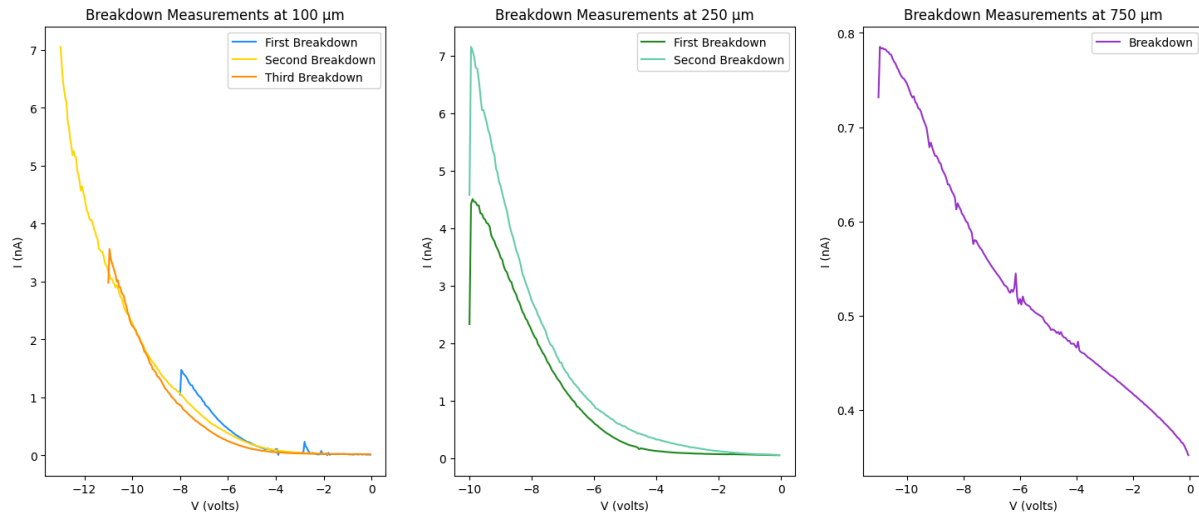
Ideality factor (Average curve): $n \approx 1.263$

Ideality factor (BR 100 μm diode): $n \approx 1.173$

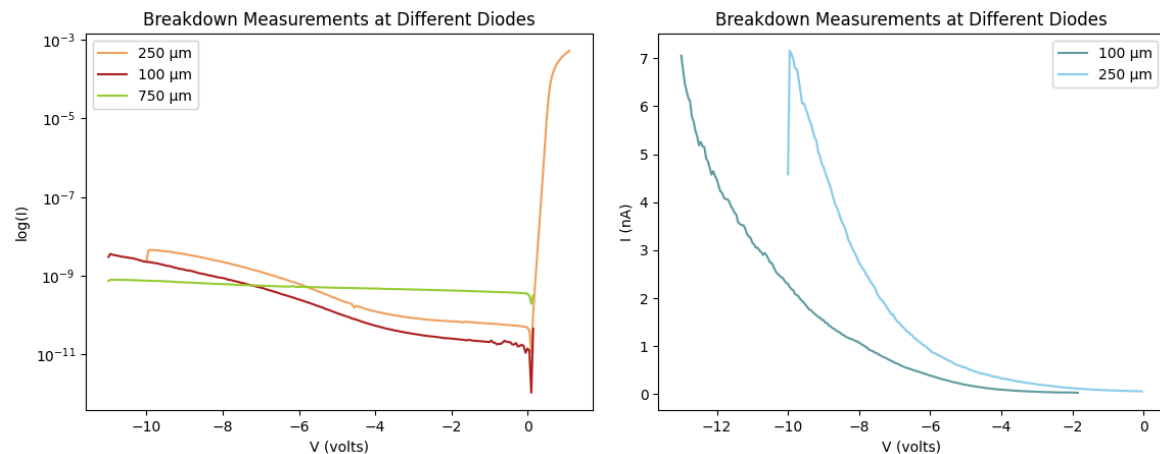
We also calculated the ideality for the Bottom Right 100 μm , which was found to be the most ideal of the diodes tested, either because of minimal defects in fabrication or improved probe contact when testing. While the Bottom Right 100 μm diode showed a slightly lower ideality factor than the average diode, the difference was modest. This indicates that the diodes generally all showed similar behavior, with current largely governed by diffusion rather than recombination or other non-ideal mechanisms.

Breakdown

We applied a greater reverse voltage on a number of diodes in order to observe breakdown. The results are plotted below.



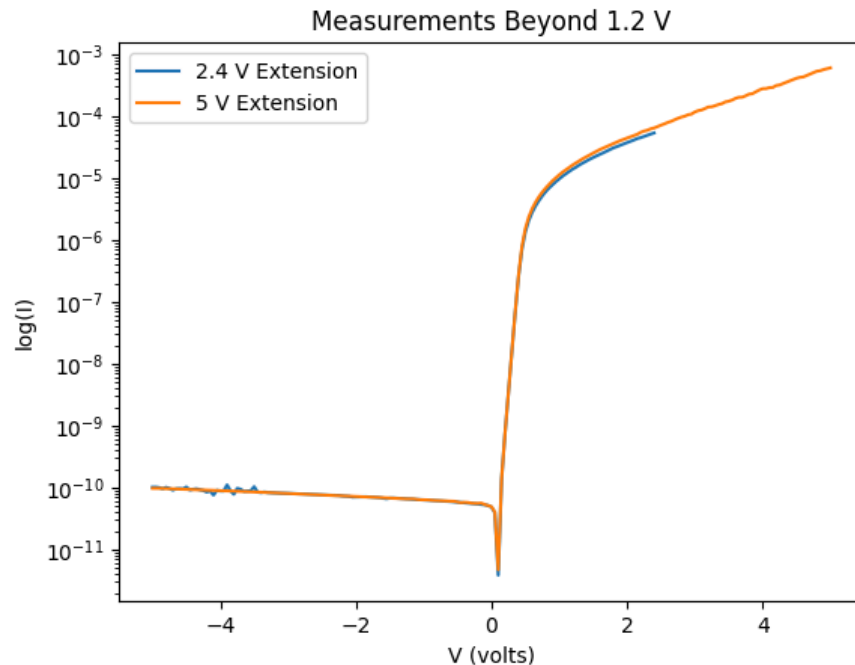
While we expected a rapid increase in reverse current at breakdown, we instead observed the gradual rise in reverse current seen above. This rise starts around -8 to -4 V and simply increases smoothly. This behavior is somewhat consistent with Zener breakdown, particularly tunneling breakdown. This is when the electric field becomes strong enough to allow quantum tunneling of electrons across the depletion region. This rise is particularly visible in the 100 and 250 μm diodes. However, if this is tunneling breakdown, it is never reaches the exponential jump in current that would indicate V_{BR} . The 750 μm exhibits an almost-linear rise, which may indicate that it never reached a large enough electric field to allow breakdown to begin occurring.



On the left, we see a log-scale comparison of the three diodes. The 250 μm diode shows the steepest rise in reverse current between -7 V and -9 V, followed by a sudden drop, likely due to current compliance limits or the diode burning out. The 100 μm diode follows a similar trend with a more gradual slope and also drops off after -10 V. The 750 μm diode remains mostly flat, indicating it likely never reached a strong enough electric field for breakdown. This supports our earlier reasoning that the 100 μm and 250 μm diodes were nearing tunneling breakdown, while the 750 μm diode did not.

The right-hand plot shows the same data on a linear scale, highlighting how rapidly current rises for the 250 μm diode. Its sharp increase and abrupt cutoff suggest it was closest to breakdown. The 100 μm curve is more modest, and again, the 750 μm shows little change. The lack of full breakdown is likely due to testing caution, as we avoided applying too large a bias to prevent blowing out the junctions. There may have also been issues with fabrication. If the depletion region was too wide in the 750 μm diode due to lighter doping or a shallower junction, the electric field may have remained below the threshold required for tunneling.

Measurements Beyond 1.2 V



We extended our forward-bias sweep up to 2.4 V and 5 V to investigate high-bias behavior. Both curves show a sharp increase in current beginning around 0.7-0.8 V, followed by a noticeable flattening in the log-scale plot at higher voltages.

This flattening indicates a departure from ideal exponential behavior and reflects non-Ohmic effects, primarily due to series resistance in the diode. As the forward current increases, voltage drops across the bulk semiconductor and contacts become significant, reducing the effective voltage across the pn junction and causing the curve to deviate from the ideal slope predicted by the diode equation. These Ohmic effects become more dominant as we approach and exceed 1.2 V. In practice, this means current no longer increases exponentially with voltage but instead becomes limited by resistive elements in the device structure.

Economic Analysis

From the above plots and analysis, we see that all the diodes we manufactured were able to deliver a forward bias current of 1mA at 1.2 Volts. We also see that at 1.2 Volts, the current rectification ratio was above the 10^{-3} minimum spec.

However, as noted above, the diodes we manufactured in the bottom left quadrant were inconsistent with the diodes on other parts of the wafer. Thus, we take our yield to be 75%. With this information, we calculate the number of manufactured devices we are able to sell in Table 1. We then use Table 2 to calculate the cost of our manufacturing process.

If we aim to make a profit margin of 20%, our total sales should be around \$432. Thus we propose the pricing structure shown in Table 3.

Table 1: Quantity of each device eligible for sale			
Size of Device (um)	Quantity Made	Yield	Quantity for Sale
750	109	0.75	81
250	218	0.75	163
100	436	0.75	327

Table 3: Pricing Structure of Devices			
Size of Device (um)	Quantity for sale	Unit Price (\$)	Total Sales (\$)
750	109	0.17	18.53
250	218	0.50	109
100	436	0.70	305.2
		Total:	\$432.73

Table 2: Total Cost to Manufacture all Devices			
Product/Service	Cost	Quantity Used	Incurred Cost
Wafer	\$30	1	\$30
Simple Solvent Clean	\$15	3	\$45
Spin Coating of Photoresist/Dopant	\$15	2	\$30
Furnace Anneal	\$15	1	\$15
HF Treatment for Oxide Removal	\$15	1	\$15
Metal Evaporation	\$70 for Gold	1	\$70
Surface Profilometry	\$5	1	\$5
Low Temp Oxide Deposition	\$50	0	\$50
Optical Lithography	\$50	2	\$100
		Total:	\$360